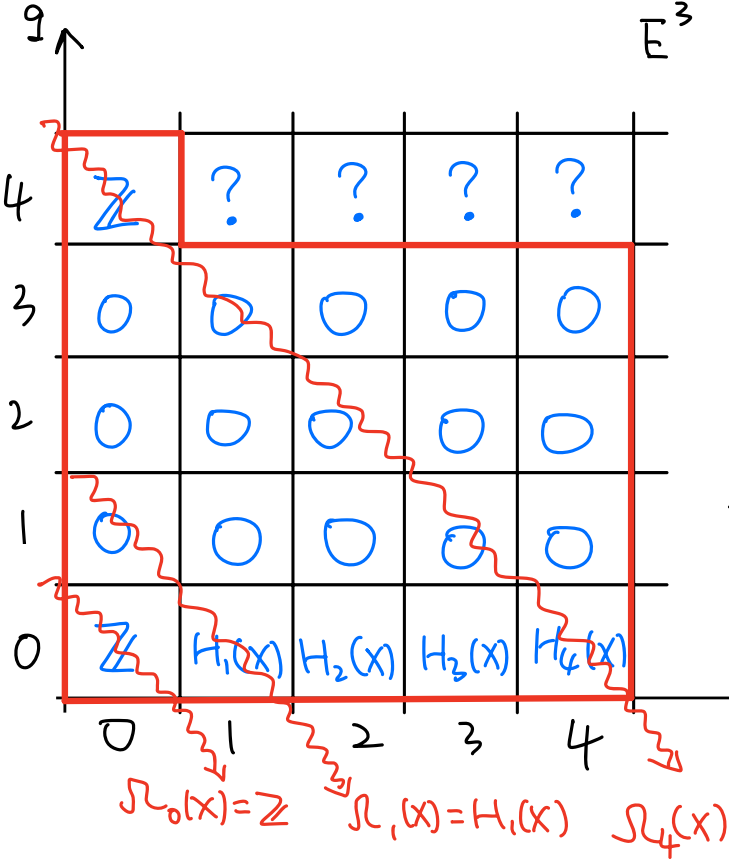




$$\begin{aligned}
 0 &\rightarrow F^1 \Omega_{\mathcal{V}}(X) \rightarrow \mathbb{Z} \\
 F^1 \Omega_{\mathcal{V}}(X) &\xrightarrow{\sim} F^2 \Omega_{\mathcal{V}}(X) \rightarrow 0 \\
 F^2 \Omega_{\mathcal{V}}(X) &\xrightarrow{\sim} F^3 \Omega_{\mathcal{V}}(X) \rightarrow 0 \\
 F^3 \Omega_{\mathcal{V}}(X) &\xrightarrow{\sim} F^4 \Omega_{\mathcal{V}}(X) \rightarrow 0 \\
 F^4 \Omega_{\mathcal{V}}(X) &\rightarrow \Omega_{\mathcal{V}}(X) \rightarrow H_4(X)
 \end{aligned}$$

Red arrows indicate the filtration steps:

- $\mathbb{Z}$ (at $p=0, q=0$)
- $\mathbb{Z} \oplus H_4(X)$ (at $p=4, q=0$)